

Total No. of Questions : 5]

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M.C.A. (Management)

MT-21-OTE-552 MJ : OPTIMIZATION TECHNIQUES
(2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Use of statistical table and non-programmable calculator is allowed.
- 3) Figures to right indicate full marks.

Q1) A) Solve the following (Any One) : [8]

- i) A furniture manufacturing company plans to make two products - chairs and tables from its available resources which consists of 400 board feet of mahogany timber and 450 manhours of labor. It knows that to make a chair requires 5 board feet and 10 man-hours and charges a cost of Rs.45/-, while each table uses 20 board feet and 15 man-hours and charges a cost of Rs. 80/- The problem is to determine how many chairs and tables the company can make keeping within its resources constraints so that it minimizes the cost. Formulate a linear programming model and find an initial basic feasible solution.
- ii) A company produces three kinds of dolls A, B and C. Their monthly production is 1000 units, 2000 units and 3000 units respectively. The dolls are sold through departmental stores X, Y and Z. 1500 units are to be supplied to each of the stores every month.

However, store Z does not want any doll of type A. Profit per unit on the dolls sold to each of the stores is given below.

	Stores		
DOLLS	X	Y	Z
A	15	10	0
B	16	8	9
C	12	9	11

Suggest optimum policy schedule and find the relevant profit.

P.T.O.

B) Explain the following (Any One) : [2]

- i) Optimum Solution
- ii) Feasible Solution in LPP

Q2) A) Solve the following (Any One) : [5]

- i) A manufacturing company processes 6 different jobs on two machines A and B. Number of units of each job and its processing times on A and B are given below. Find the optimal sequence, the Total minimum elapsed time and idle time for each machine

Job	Number of Units of each job	Processing time Machine A	Processing time Machine B
1	3	5	8
2	1	6	4
3	2	6	11
4	4	8	7
5	1	9	12
6	2	6	14

- ii) Find the sequence that minimizes the total time required in performing the following job on three machines in the order BAC. Processing times (in hours) are given in the following table.

Job	1	2	3	4	5
Machine A	5	6	2	3	4
Machine B	7	5	6	7	4
Machine C	8	9	8	7	6

Also find Total elapse time and Idle time for each machine.

B) Solve the following (Any One) : [5]

- i) A bank operates with a single teller and experiences customer arrivals following a Poisson distribution with an average of 12 customers per hour. The teller takes on an average 4 minutes to serve each customer.
 - a) Calculate the utilization factor of the system.
 - b) What is the probability that there are 0 customers in the system (i.e., the system is empty)?
 - c) Compute the expected number of customers in the system at any given time.

- ii) Arrivals at a telephone both are considered to be Poisson at an average time of 6 min between one arrival and the next. The length of the phone call is distributed exponentially, with a mean of 5 min.

Determine :

- Expected waiting time in the queue.
- Expected waiting time in the system.
- Expected number of units in queue that from time to time.

Q3) A) Solve the following (Any One) : [5]

- i) Three companies, A, B and C, compete against each other. The transition matrix T for people switching each month among them is given by the following transition matrix.

	A	B	C
A	0.1	0.3	0.6
B	0.2	0.6	0.2
C	0.10	0.25	0.65

Find the long term distribution?

- ii) In certain market, only two brands of ice creams A, B are sold. Given that a man has last purchased brand A, there is 80% chance that he would buy brand B in next purchase. While if man purchase brand B, there is 90% chance that he would buy same brand A in next purchase.

- Develop transition probability matrix.
- Draw transition diagram.

B) Solve the following (Any One) : [5]

- i) The daily demand for Milk bottles, based on past experience, is given as follows:

Daily Demand	10	20	30	40	50
Cumulative Probability	0.10	0.40	0.65	0.85	1.00

A manager decides to keep 45 Milk bottles in stock. Using the following random numbers: 69, 45, 58, 74, 82, 25, 72, 44, 75, 12, estimate the average balance stock.

- ii) Over 130 days' period, the daily demand of a certain commodity shows the following frequency distribution pattern.

Simulate the demand for 10 days using the random numbers 67, 84, 01, 77, 90, 14, 15, 74, 44, 77. Also find the average demand

Daily Demand	1	2	3	4	5
No. of days	26	13	39	13	39

Q4) A) Solve the following (Any One) :

[8]

- i) Consider a CPM network with the following project data:

Activity	Duration (days)	Predecessor(s)
A	4	None
B	7	None
C	6	B
D	5	B
E	12	A
F	4	A
G	4	F
H	6	C, E, G
I	2	F

- Draw a network diagram
 - Calculate Critical Path (Perform Forward Pass and Backward Pass calculations)
 - Calculate Free Float for each activity
- ii) Given the following information

Activities	1-2	1-3	1-4	2-6	3-5	4-5	4-7	5-6	5-7	6-8	7-8
Optimistic Time	6	1	1	1	1	1	2	4	4	2	2
Pessimistic Time	8	9	7	3	9	9	8	4	10	14	8
Most Likely Time	7	2	4	2	2	5	2	4	4	5	2

- Construct the project network
- Find the critical path (Perform Forward Pass and Backward Pass calculations) and project variance
- If the probability of completing the project is 0.84, find the expected project completion time?

B) Explain the following (Any One) : [2]

- i) Critical Path
- ii) Earliest Finish Time and Latest Start Time

Q5) A) Solve the following (Any One) : [5]

- i) A grocery receives its weekly supply of eggs every Thursday morning. This shipment must last until the following Thursday when a new shipment is received. Any eggs left unsold by Thursday are destroyed. Eggs sell for \$10 per hundred and cost \$7 per hundred.

The weekly demand for eggs at this grocery varies from week to week. From past experience, the following probability distribution is assigned to weekly demand:

Demand (hundreds of eggs):	11	12	13	14	15
Probability:	0.1	0.2	0.3	0.3	0.1

This pattern of demand remains stable throughout the year. The demand for eggs is not seasonal, and the trend is flat. The problem is: How many eggs should be ordered for delivery every Thursday?

- ii) The following matrix gives the payoff of different conditions (events) E1, E2, E3 and E4 against strategies (alternatives) (A1, A2, A3, A4). Indicate the decision under regret approach if $P(E1) = 0.25$, $P(E2) = 0.35$, $P(E3) = 0.25$, $P(E4) = 0.15$

Find the regret table from the following pay-off table and suggest optimum strategy.

Events / State of Nature / Outcomes	Actions / Strategies			
	A1	A2	A3	A4
E1	20	40	-20	60
E2	30	50	20	320
E3	-20	10	30	310
E4	60	30	110	45

B) Solve the following (Any One) :

[5]

i) Find the optimum strategies of the player in the following game, also find

- Value of game
- Optimum strategy for Player A
- Optimum Strategy for Player B

Player A	Player B			
		B 1	B 2	B 3
	A1	30	40	-80
	A2	0	50	-20
	A3	90	20	50

ii) Find the optimum strategies of the player in the following game, also find

- Value of game
- Optimum strategy for Player A
- Optimum Strategy for Player B

A	B			
		B 1	B 2	B 3
	A1	7	8	5
	A2	6	10	-6
	A3	5	-5	4
